

AMPHIBIANS

PHYLUM	Chordata
CLASS	Amphibia
ORDERS	3
FAMILIES	75
SPECIES	7,615

Amphibians are intimately associated with water, most of them spending part of their life in water and part on land. These ectothermic (cold-blooded) vertebrates are divided into 3 groups: newts and salamanders; frogs and toads; and the

wormlike caecilians. The life cycle of most amphibians involves a transformation from aquatic larvae (which breathe using gills) to terrestrial adults (which take in oxygen through lungs). Most live close to water in tropical and temperate regions, although some have adapted to survive cold and drought.



TREE FROG
This tree frog, from the rain forests of New Guinea and northern Australia, has moist skin that is well suited to its wet and humid habitat.

Past and present

The first amphibians, which appeared about 370 million years ago during the Devonian Period, were descendants of fleshy-finned fish. They were the first vertebrates to colonize the land, and had scaly skin and a finned tail, but also lungs that could breathe air. They were also the first quadrupeds, evolving 2 pairs of jointed limbs from the fins of



AMPHIBIAN ANCESTOR
Diplocaulus lived about 280 million years ago and probably resembled a modern newt, but for an unusually shaped head.

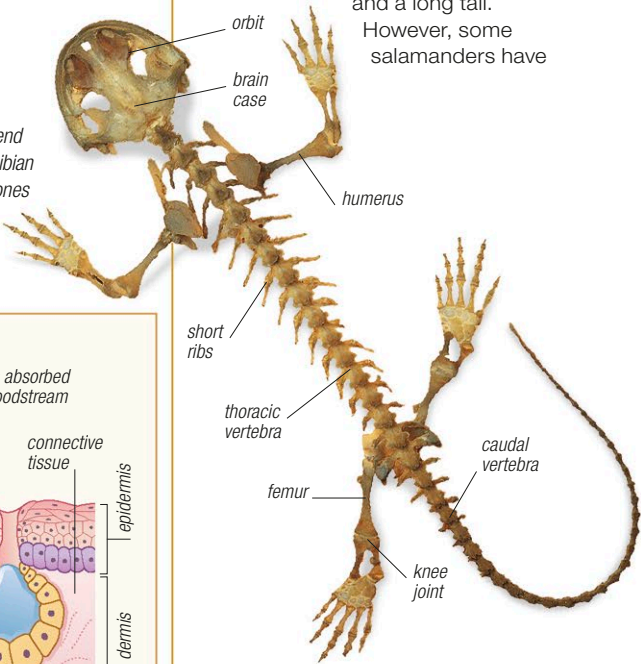
their fish ancestors. These enabled them to lift their bodies off the ground, although they were still heavily built animals, with slow and clumsy movements compared to amphibians alive today. By about 280 million years ago, amphibians had become the dominant land animals. However, over the next 70 million years, amphibians were superseded by terrestrial reptiles (which had evolved the ability to lay shell-covered eggs on land) and most species gradually died out.

Much more recently, amphibians have been struck by a far more rapid decline. Many living species are endangered—nearly 500 critically so—and in the last 2 decades, a number

have become extinct. The reasons for this are complex, and not fully understood, but they probably involve a range of factors, acting both on a local and global scale. Among them are contamination by waterborne pollutants, infection by chytrid fungi, and climate change. Chytrid disease was first observed in Australia, but it now affects amphibians on several continents, and may eventually spread worldwide.

SKELETON

This skeleton of a juvenile Japanese giant salamander highlights an evolutionary trend toward simplifying the amphibian skeleton. It contains fewer bones than other vertebrates, and far fewer than its fish ancestors.



Anatomy

Compared to other vertebrates, amphibians have a fairly simple body structure, well adapted for life in water and on land. Most adult amphibians have 4 limbs, with 4 digits on the front limbs and 5 on the back. Newts and salamanders have a relatively small head, a long, slender body, with short limbs of equal length, and a long tail. However, some salamanders have

SKIN

The skin of an amphibian is naked and smooth, with no scales or hair to protect it. It is permeable to water and, although numerous mucous glands help to keep the skin moist, most amphibians quickly dry out if they are not in a damp place. All amphibians have poison glands in their skin, and many produce secretions that are distasteful or highly toxic to potential predators. The skin usually contains numerous pigment cells, and many amphibians are brightly colored. This is especially common in poisonous species, in which striking skin patterns serve as a warning. Many amphibians use their skin for breathing, which



IN SUNLIGHT

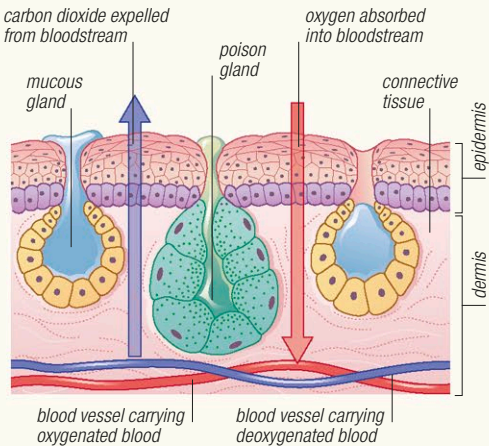


IN SHADE

means that it has to be kept moist at all times. This may also place a limit on how large amphibians can be, since they need to maintain a high ratio of surface area to volume, to absorb an adequate amount of oxygen.

CHANGING SKIN COLOR

The base color of some amphibians' skin changes in response to variations in temperature, light level, or mood. When in sunlight, the skin of the Australian green tree frog (left) is light green but on entering shade gradually turns brown.



BREATHING THROUGH THE SKIN

Many amphibians can breathe through their skin, absorbing oxygen directly into their bloodstream and expelling carbon dioxide. In addition to being packed with blood vessels—some only a few cells' width from the surface—the skin is very thin, enabling the easy transfer of gases. Mucous glands keep the surface damp, helping the gases to pass through.

very small limbs with reduced numbers of digits, or lack rear limbs entirely. Most frogs and toads have a distinctive compact body shape, with a large head, short back, small front limbs, large, muscular back limbs, and no tail. Caecilians have a wormlike, limbless body, with a pointed head and a short tail. Other than caecilians, which use smell, most amphibians hunt using sight and have large eyes that help them to locate prey at night. They